Ziyang Song

Contact Information

Stocker Center 313
School of Electrical Engineering and Computer Science
Ohio University
Athens, Ohio, United States

ziyangs@ohio.edu
Website
LinkedIn
Google Scholar

Research Interests

My research focuses on AI for health, with a focus on generative AI and statistical machine learning. My vision is to build trustworthy AI to address biomedical challenges, including:

1. **Trustworthy LLMs and Agents in Healthcare.** I develop methods to improve **explainability**, **uncertainty quantification**, and **prediction safety** for biomedical LLMs and multi-agent clinical decision-support systems.
2. **Generative AI for Health Time Series.** I design representation learning and generative modeling techniques for irregular, longitudinal clinical data, including electronic medical records, laboratory measurements, and biosignals, enabling forecasting and health trajectory analysis.
3. **Generative AI for Biology.** I build generative models for single-cell RNA-seq and spatial omics to infer cellular trajectories and characterize spatiotemporal developmental patterns.
4. **Statistical Machine Learning for Medical NLP.** I develop probabilistic models and scalable statistical inference to learn interpretable medical representations from electronic health records and genotype data.

Full-time Professional Experience

Ohio University — Athens, OH, United States
School of Electrical Engineering and Computer Science
Assistant Professor — Aug. 2025 – Present

Education

McGill University — Montreal, QC, Canada
Ph.D. in Computer Science — Sep. 2020 – May 2025
- **Affiliation:** Mila – Quebec AI Institute
- **Advisor:** Yue Li

Concordia University — Montreal, QC, Canada
M.Sc. in Quality Systems Engineering — Sep. 2018 – Nov. 2019
- **Advisor:** Nizar Bouguila

University of Shanghai for Science and Technology — Shanghai, China
B.Eng. in Computer Science — Sep. 2014 – Jun. 2018

Publications

Note: † equal contribution; * corresponding authors.

Conference Papers:

1. **Ziyang Song**, Qincheng Lu, Hao Xu, He Zhu, David Buckeridge, Yue Li. TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *ACM-BCB 2024*. **Rising Star Award**.
2. Zilong Wang, Ruohan Wang, **Ziyang Song**, David Buckeridge, Yue Li. MixEHR-Nest: Identifying subphenotypes within electronic health records through hierarchical guided-topic modeling. *ACM-BCB 2024*.
3. **Ziyang Song**, Qincheng Lu, He Zhu, David L. Buckeridge, Yue Li. Bidirectional generative pre-training for improving healthcare time-series representation learning. *MLHC (PMLR) 2024*.
4. **Ziyang Song**, Yuanyi Hu, Aman Verma, David L. Buckeridge, Yue Li. Automatic phenotyping by a seed-guided topic model. *KDD 2022*. **Health Day Best Paper Award**.

5. **Ziyang Song**, Xavier Sumba Toral, Yixin Xu, Aihua Liu, Liming Guo, Guido Powell, Aman Verma, David Buckeridge, Ariane Marelli, Yue Li. Supervised multi-specialist topic model with applications on large-scale electronic health record data. *ACM-BCB 2021.*
6. **Ziyang Song**, Ornela Bregu, Samr Ali, Nizar Bouguila. Variational inference of finite asymmetric gaussian mixture models. *IEEE SSCI 2019.*
7. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction. *ICIAR 2019.*

Journal Papers:

1. Ziqi Yang[†], **Ziyang Song**[†], Shadi Zabad, Marc-André Legault, Yue Li. PheCode-guided multi-modal topic modeling of electronic health records improves disease incidence prediction and GWAS discovery from UK Biobank. *Briefings in Bioinformatics.* (2026).
2. **Ziyang Song**, Qincheng Lu, He Zhu, David Buckeridge, Yue Li. TrajGPT: Irregular Time-Series Representation Learning of Health Trajectory. *IEEE Journal of Biomedical and Health Informatics (J-BHI).* (2025).
3. **Ziyang Song**, Qincheng Lu, Hao Xu, Ziqi Yang, He Zhu, David Buckeridge, Yue Li. TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *Health Information Science and Systems.* (2025).
4. Yuesong Zou, Ahmad Pesaranghader, **Ziyang Song**, Aman Verma, David Buckeridge, Yue Li. Modeling electronic health record data using an end-to-end knowledge-graph-informed topic model. *Scientific Reports.* (2022).
5. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian inference for infinite asymmetric gaussian mixture with feature selection. *Soft Computing.* (2021).
6. **Ziyang Song**, Samr Ali, Nizar Bouguila. Background subtraction using infinite asymmetric Gaussian mixture models with simultaneous feature selection. *IET Image Processing.* (2020).
7. **Ziyang Song**, Samr Ali, Nizar Bouguila, Wentao Fan. Nonparametric hierarchical mixture models based on asymmetric gaussian distribution. *Digital Signal Processing.* (2020).

Preprints & Under Submission:

1. **Ziyang Song**[*], Jiahe Shen, Yixuan Li, Qincheng Lu, Weihao Li, Haoran Xu, He Zhu, Xinyu Wang, Yue Li. SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation. *NeurIPS Responsible Foundation Model Workshop.* Submitted to *ICML 2026*, under review.
2. Xinyu Wang, Yixuan Li, Hanwei Wu, Yixuan Li, Qiguang Ye, **Ziyang Song**[*]. Patients-like-me: Semantic Medical ID for Hierarchy-Aware Concept Representation. Submitted to *ICML 2026*, under review.
3. Ruilin Wang, Bo-Hong Wang, Elizabeth Kourbatski, Hegang Chen, **Ziyang Song**, Gilles Boire, Marie Hudson, Yue Li. SciKit-Agent: Iterative Agentic Refinement for Tabular ML Pipelines. Submitted to *ICML 2026*, under review.
4. Jun Bai, **Ziyang Song**, Yue Li. FedEHR-GEN: Federated Synthetic Time-Series EHR Generation via Latent Space Alignment and Distribution-Aware Aggregation. Submitted to *KDD 2026*, under review.

Invited Talks & Presentations

Invited Talks

Northern Arizona University	*Representation Learning of Electronic Health Records.*	March. 2025
San José State University	*Representation Learning of Electronic Health Records.*	March. 2025
Ohio University	*Representation Learning of Electronic Health Records.*	Feb. 2025
South Dakota State University	*Representation Learning of Electronic Health Records.*	Feb. 2025
University of Texas at El Paso	*Representation Learning of Electronic Health Records.*	Feb. 2025
Microsoft Research Asia (MSRA)	*Time-series Transformer for Representation Learning and Health Forecasting*	Nov. 2024

Guest Lectures

Ohio University	*Generative AI for Healthcare.*	Nov. 2025
McGill University	*Time-series Transformer in Electronic Health Records.*	Nov. 2024
McGill University	*Time-series Transformer in Electronic Health Records.*	Nov. 2023

Oral Presentations

Canadian Society for Epidemiology and Biostatistics (CSEB)	*AI-driven approach for computational disease phenotyping within the CARTaGENE cohort..*	Aug. 2025
ACM BCB	*TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare.*	Nov. 2024
Scientific Meeting of the Canadian Translational Geroscience Network	*AI-driven approach for computational phenotyping with CARTaGENE cohort.*	Sep. 2024
Tokyo Symposium on Genomic Medicine, Therapeutics and Health	*Tutorial: Phenotyping and PheWAS using MixEHR-Seed.*	Apr. 2024
ISMB GLBIO	*MixEHR-SAGE: A seed-guided topic model to improve phenotyping for PheWAS analysis in UK Biobank data.*	May. 2023
KDD	*Automatic phenotyping by a seed-guided topic model.*	Aug. 2022
ACM BCB	*Supervised multi-specialist topic model with applications on large-scale electronic health record data.*	Aug. 2021

Poster Presentations

NeurIPS 2025 Responsible Foundation Model Workshop	*SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation.*	Nov. 2025
NeurIPS 2024 TSALM Workshop	*TrajGPT: Healthcare Time-Series Representation Learning for Trajectory Prediction.*	Dec. 2024
MLHC	*Bidirectional generative pre-training for improving healthcare time-series representation learning.*	Aug. 2024
50th Anniversary of SOCS at McGill University	*MixEHR-Seed: Automatic phenotyping by a seed-guided topic model.*	Oct. 2022
ICIAR	*Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction.*	Aug. 2019

Awards, Grants, and Funding

As Principal Investigator (August 2025 onwards)

NVIDIA Academic Grant Program Award (PI)	*Gift*	Dec. 2025

Prior to Principal Investigator Role (Before August 2025)

FRQNT Provincial Doctoral Research Scholarship	*Scholarship*	May. 2023–Sep. 2025
ACM BCB Rising Star Award	*Award*	Nov. 2024
KDD Health Day Best Paper Award	*Award*	Aug. 2022

Teaching Experience

Ohio University — Athens, OH, United States

Instructor — Spring 2026
AI 3300 Statistical Learning

Instructor — Fall 2025
AI 5010/4010 Foundations of Deep Learning

McGill University — Montreal, QC, Canada

Teaching Assistant — Winter 2023
COMP 551 Applied Machine Learning

Teaching Assistant — Fall 2022
COMP 565 Machine Learning in Genomics and Healthcare

Internship Experience

Centre Hospitalier Universitaire Sainte-Justine (CHU-SJ) — Montreal, QC, Canada
Research Intern — Jun. 2023–Dec. 2024
Developed a clinical decision-support system using a probabilistic model to identify high-risk patients who may be missed by existing rule-based screening methods.

Nanyang Technological University — Singapore
International Research Intern — Jan. 2020–Jun. 2020
Developed probabilistic models with stochastic optimization to analyze retail data, improving personalized product recommendations and targeted promotions.

Supervision & Mentoring

Ohio University

Ph.D. Students

Long Nguyen Tan — Jan. 2026–Present

Research/RA Students

Hanwei Wu (M.Sc., McMaster University) — May. 2025–Present
Mentored in fine-tuning LLMs and graph machine learning.

Jiahe Shen (M.Sc., UW–Madison) — Jul. 2025–Present
Mentored in applying semantic IDs in biomedical LLMs; submitted to ICML 2026.

Zihao Li (M.Sc., Northwestern University) — Jul. 2025–Present
Mentored in representation learning of medical concepts in hyperbolic spaces.

Yiran Ding (M.Sc., Brown University) — Jul. 2025–Present
Mentored in contrastive learning and optimal transport for biomedical LLMs.

Xiaohang Xie (B.S., OSU) — Jul. 2025–Present
Mentored in confidence calibration for clinical reasoning and LLMs.

Yang Hu (B.S., OSU) — Jul. 2025–Present
Mentored in prediction confidence and rejection for LLMs.

Zhongning Deng (B.S., OSU) — Jul. 2025–Present
Mentored in uncertainty quantification for LLM outputs.

McGill University

Ruilin Wang (M.Sc., McGill University) — Sep. 2024–May. 2025
Mentored in applying LLMs to healthcare time-series data (e.g., biosignals).

Bo Hong Wang (M.Sc., McGill University) — Sep. 2024–May. 2025
Mentored in applying LLMs to healthcare time-series data (e.g., ICU lab tests).

Ziqi Yang (B.S./M.Sc., McGill University) — Sep. 2022–May. 2025
Mentored in applying MixEHR-Seed on UK Biobank; presented at ISMB GLBIO 2023.
Placement: M.Sc. student, McGill University.

Ruohan Wang (B.S./M.Sc., McGill University) — Sep. 2022–Sep. 2024
Mentored in designing MixEHR-Nest on Quebec PopHR; published at ACM-BCB 2024.
Placement: Ph.D. student, Brown University.

Hao Xu (B.S., McGill University) — May. 2023–Sep. 2023
Mentored in applying TimelyGPT on biosignals; published at ACM-BCB 2024.
Placement: Ph.D. student, University of Pennsylvania.

Ziyu Zhao (B.S., McGill University) — May. 2023–Sep. 2023
Mentored in applying TimelyGPT on Quebec PopHR database.
Placement: M.Sc. student, McGill University.

Yuanyi Hu (B.S., McGill University) — May. 2021–May. 2022
Mentored in designing MixEHR-Seed; published at KDD 2022.
Placement: M.Sc. student, Columbia University.

Yixin Xu (B.S., McGill University) — Sep. 2020–Sep. 2021
Mentored in applying MixEHR-S; published at ACM-BCB 2021.
Placement: M.Sc. student, Duke University.

Service

Conference Reviewer

ICML	2025–2026
ICLR	2025–2026
IJCAI	2025–2026
ACM SIGSPATIAL	2025
KDD	2024–2025
NeurIPS	2024
ML4H	2024

Journal Reviewer

BMJ Digital Health & AI (expert peer reviewer)	2025–2026
TMLR	2025–2026
Applied Intelligence	2025
Big Data Mining and Analytics	2024–2025

Ohio University

CS/AI Assessment & Accreditation Committee member — 2025–2026

McGill University and Mila

Evaluation committee for Mila Supervision Request Process — 2024

Referees

Dr. Yue Li, Associate Professor, School of Computer Science, McGill University
Email: send.Li.D38D6CE548@interfoliodossier.com

Dr. David Buckeridge, Professor, School of Population and Global Health, McGill University
Email: send.Buckeridge.BC61C28AEC2@interfoliodossier.com

Dr. Archer Yi Yang, Associate Professor, Department of Mathematics and Statistics, McGill University
Email: send.Yang.BCEBBEFEC2@interfoliodossier.com